
DAISY SEED RUST GUITAR DSP

// PROJECT DOCUMENTATION v1.0 // SYSTEM STATUS: OPERATIONAL

[PROJECT GOAL]

Build a custom DSP-powered electric guitar system using a Daisy Seed microcontroller as the embedded audio processing core. The final product is a physically modified guitar with embedded real-time effects processing, programmable effect chains, custom parameter mapping via knobs and encoders, and a self-contained battery-powered audio engine – all running Rust firmware on bare metal.

[CURRENT STATUS]

The project has reached a working **end-to-end audio signal chain**: audio input → Daisy Seed DSP processing → audio output, with real-time digital gain control. A rotary encoder provides bidirectional parameter input, proving the control loop that will eventually map to effect parameters like delay time, reverb mix, and distortion gain.

[MILESTONES COMPLETED]

Session 1 // Toolchain & Flashing

Established a working Rust embedded toolchain targeting the STM32H750 inside the Daisy Seed Rev7. Resolved a critical issue where ELF files were being flashed instead of raw binaries, and fixed linker configuration (missing link.x, incorrect RAM origin at 0x20000000 corrected to 0x24000000). The build-convert-flash workflow is now reliable using cargo build, rust-objcopy, and dfu-util.

Session 2 // GPIO & Input Control

Achieved LED blink as proof of code execution, then wired a KY-040 rotary encoder to GPIO pins PC0 and PC1. Built a non-blocking polling loop that reads encoder rotation and adjusts LED blink rate in real time – the foundational pattern for all future parameter control.

Session 3 // Analog Preamp Circuit

Built an OPA2134-based preamp on a breadboard with a resistive voltage divider creating a 1.65V virtual ground for single-supply operation. Verified bias voltage with a multimeter. Connected a ceramic humbucker pickup through a 10µF coupling capacitor into the op-amp's non-inverting input. First soldering experience – soldered header pins onto the TPA6132 headphone amplifier board.

Session 4 // Audio Codec & DMA Streaming

Discovered that the Daisy Seed Rev7's PCM3060 codec operates in hardware control mode (no I2C), causing the daisy crate's board_split_audio! macro to panic. Wrote a custom SAI1 + DMA audio driver from scratch using stm32h7xx-hal, bypassing I2C initialization entirely. Configured DMA double-buffering with transmit and receive streams, placed DMA buffers in D2 SRAM at 0x30000000 for DMA1 accessibility, and achieved stable 48kHz/24-bit audio streaming. Confirmed audio passthrough by playing music from a phone through a 3.5mm aux cable into the codec input and

hearing it through the TPA6132 headphone amplifier.

[TECH STACK]

Hardware

- > Daisy Seed Rev7 (STM32H750, ARM Cortex-M7, PCM3060 audio codec)
- > OPA2134PA dual op-amp (DIP-8, input buffer/preamp)
- > TPA6132 headphone amplifier board
- > KY-040 rotary encoder modules
- > Ceramic humbucker pickups (Invader-style, neck & bridge)
- > TP4056 LiPo charger + MT3608 boost + AMS1117 3.3V regulator (future)
- > Solderless breadboard for prototyping

Firmware

- > Language: Rust (no_std, no_main, bare metal)
- > Target: thumbv7em-none-eabihf (ARM Cortex-M7 with hardware float)
- > HAL: stm32h7xx-hal v0.16 for SAI, DMA, GPIO, clock configuration
- > Runtime: cortex-m-rt v0.7 with custom memory.x linker script
- > Clock: daisy crate v0.11 (seed_1_2) for PLL config only (480MHz + PLL3 audio)
- > Audio: Custom SAI1 I2S + DMA1 double-buffered @ 48kHz/24-bit

Tools

- > IDE: Zed | OS: macOS
 - > Flashing: dfu-util via USB DFU mode
 - > Binary conversion: rust-objcopy (cargo-binutils)
 - > PCB design: KiCad (installed, future use)
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[BUILD & FLASH]

```
$ cargo build --release --target thumbv7em-none-eabihf

$ rust-objcopy -O binary --only-section=.vector_table \
  --only-section=.text --only-section=.rodata \
  target/.../daisy_blink target/.../daisy_blink.bin

$ # DFU mode: hold BOOT, press RESET, release RESET, release BOOT

$ dfu-util -a 0 -s 0x08000000:leave -D target/.../daisy_blink.bin

$ # Press RESET after flashing
```

[VERIFIED PINOUT // DAISY SEED REV7]

Orientation: USB port at bottom, pin 1 bottom-right, pin 40 bottom-left.

PIN	FUNCTION	VOLTAGE AT REST
10	3V3	3.24V

16	Audio In L	0.69V (codec bias)
17	Audio In R	0.70V (codec bias)
18	Audio Out L	0.02V
19	Audio Out R	0.02V
20	AGND	0.01V
21	3V3 Analog	3.25V
22	PC0 (GPIO/ADC)	floating
27	PC1 (GPIO/ADC)	floating
38	3V3 Digital	3.28V
39	VIN (USB 5V)	4.74V
40	DGND	0.0V

[KNOWN ISSUES & WORKAROUNDS]

PCM3060 hardware mode

The Rev7 codec does not respond to I2C. The daisy crate's `board_split_audio!` macro panics. Workaround: initialize SAI1 and DMA manually, skip I2C entirely.

DMA buffer placement

DMA1 cannot access AXI SRAM (0x24000000). Buffers must be placed in D2 SRAM (0x30000000) using raw pointer addresses.

Binary conversion overflow

`rust-objcopy` without `--only-section` flags produces a ~448MB file due to the address gap between flash (0x08000000) and SRAM (0x30000000). Always specify sections explicitly.

DFU `get_status` error

Normal STM32 DFU timing quirk, not a real failure.

[NEXT STEPS]

- > Add first DSP effect (distortion, delay, or EQ) to audio passthrough
- > Reconnect rotary encoder for real-time effect parameter control
- > Improve analog input gain staging for guitar pickup signal levels
- > Migrate from breadboard to prototype PCB
- > Design battery power system (LiPo + TP4056 + boost converter)
- > Transition from Daisy Seed to Daisy SM for final guitar hardware
- > Custom PCB design in KiCad

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